

Real Analysis HW 4

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BS - Page 67 Exercise 13

Use the Squeeze Theorem to determine the limits of

- $\left(n^{\frac{1}{n^2}}\right)$
- $\left((n!)^{\frac{1}{n^2}}\right)$

(a)

$$n^{\frac{1}{n^2}} = e^{\ln(n^{\frac{1}{n^2}})} = e^{\frac{\ln(n)}{n^2}}$$

Now we only need to consider $\frac{\ln(n)}{n^2}$

Since $\ln(n) \geq 0 \forall n$ and $\ln(n) \leq n \forall n$, proofs at the end of this question, we have

$$0 \leq \frac{\ln(n)}{n^2} \leq \frac{n}{n^2} = \frac{1}{n}$$

Then by Squeeze Theorem, since $\frac{1}{n} \rightarrow 0$, we have that $\frac{\ln(n)}{n^2}$ converges to 0

Thus $e^{\frac{\ln(n)}{n^2}} = n^{\frac{1}{n^2}}$ converges to $e^0 = 1$

(b)

Similarly for the second limit we get,

$$(n!)^{\frac{1}{n^2}} = e^{\frac{\ln(n!)}{n^2}}$$

$$\frac{\ln(n!)}{n^2} = \frac{1}{n^2}(\ln(n) + \ln(n-1) + \dots + \ln(1))$$

Then since $\ln(n) \leq \ln(n+1) \forall n$ we get,

$$0 \leq \frac{\ln(n!)}{n^2} \leq \frac{1}{n^2}(n \ln(n))$$

$$0 \leq \frac{\ln(n!)}{n^2} \leq \frac{1}{n}(\ln(n))$$

We have $\ln(n) \leq n \Rightarrow \ln(\sqrt{n}) \leq \sqrt{n} \Rightarrow \ln(n) \leq 2\sqrt{n}$

$$0 \leq \frac{\ln(n!)}{n^2} \leq \frac{2\sqrt{n}}{n} = \frac{2}{\sqrt{n}}$$

As $n \rightarrow \infty$, $\sqrt{n} \rightarrow \infty$, since we know $\frac{1}{n} \rightarrow 0$ then $\frac{2}{\sqrt{n}} \rightarrow 0$

Thus $\frac{\ln(n!)}{n^2}$ too converges to 0, by Squeeze Theorem

Thus $e^{\frac{\ln(n!)}{n^2}} = (n!)^{\frac{1}{n^2}}$ converges to $e^0 = 1$

Proofs of properties used

- I will prove $\ln(n) \geq 0 \forall n$ by induction,

Base Case:

$$\ln(1) = 0 \geq 0$$

Induction Step:

Assume it holds for $n = k$

$$\ln(k) \geq 0$$

$$k \geq e^0 = 1$$

$$k + 1 \geq 2$$

$$\ln(k + 1) \geq \ln(2) \geq 0$$

Thus $\ln(n) \geq 0 \forall n$

- $\ln(n) \leq n \forall n$ will also be proven by induction

Base Case:

$$\ln(1) = 0 \leq 1$$

Induction Step:

Assume it holds for $n = k$

$$\ln(k) \leq k$$

$$k \leq e^k$$

$$ke \leq e^{k+1}$$

Also $k + 1 \leq ke \forall k$

– Base Case:

$$2 \leq e$$

Induction Step:

Assume it holds for $k = m$

$$m + 1 \leq me$$

$$m + 1 + e \leq e(m + 1)$$

$$m + 2 \leq m + 1 + e \leq e(m + 1)$$

Then

$$k + 1 \leq ke \leq e^{k+1}$$

$$\ln(k + 1) \leq k + 1$$

Thus by induction, $\ln(n) \leq n \forall n$

BS - Page 80 Exercise 5

Let $X = \{x_n\}$ and $Y = \{y_n\}$ be given sequences and let the "shuffled" sequence $Z = \{z_n\}$ be defined by $z_1 = x_1, z_2 = y_1, \dots, z_{2n-1} = x_n, z_{2n} = y_n, \dots$

Show that Z is convergent iff both X and Y are convergent and $\lim X = \lim Y$

($\Leftarrow$)

Assume $\{x_n\}, \{y_n\}$ convergent, both with limit L

$$\Rightarrow \forall \epsilon > 0 \exists N_1, N_2 \in \mathbb{N} \text{ such that}$$

- $|x_n - L| < \epsilon \forall n \geq N_1$

- $|y_n - L| < \epsilon \forall n \geq N_2$

$\{z_n\}$ convergent with limit z if $\forall \epsilon > 0 \exists N_3 \in \mathbb{N}$ such that $|z_n - z| < \epsilon \forall n \geq N_3$

There are two cases for each z_n

$$z_n = \begin{cases} x_{\frac{n+1}{2}}, & n \text{ odd} \\ y_{\frac{n}{2}}, & n \text{ even} \end{cases}$$

$\forall n \geq \max\{N_1, N_2\}$, we have $|x_n - L| < \epsilon$ and $|y_n - L| < \epsilon$

$\Rightarrow \forall n \geq \max\{2N_1, 2N_2\}$, we have $|x_{\frac{n+1}{2}} - L| < \epsilon$ if n is odd and $|y_{\frac{n}{2}} - L| < \epsilon$ if n is even.

$\Rightarrow \forall \epsilon > 0 \exists N_3 \in \mathbb{N}$ such that $|z_n - L| < \epsilon \forall n \geq N_3$

$\Rightarrow \{z_n\}$ convergent with limit L

($\Rightarrow$)

Assume $\{z_n\}$ convergent with limit L

$\forall \epsilon > 0 \exists N_1 \in \mathbb{N}$ such that $|z_n - L| < \epsilon \forall n \geq N_1$

There are two cases for each z_n

$$z_n = \begin{cases} x_{\frac{n+1}{2}}, & n \text{ odd} \\ y_{\frac{n}{2}}, & n \text{ even} \end{cases}$$

$\forall n \geq N_1$, of the form $n = 2k + 1, k \in \mathbb{N} \Rightarrow |x_{k+1} - L| < \epsilon \forall k \geq \frac{N_1-1}{2}$

$\forall n \geq N_1$, of the form $n = 2k, k \in \mathbb{N} \Rightarrow |y_k - L| < \epsilon \forall k \geq \frac{N_1}{2}$

$\Rightarrow \forall \epsilon > 0 \exists N_2 \in \mathbb{N}$ such that

- $|x_n - L| < \epsilon \forall n \geq N_2$
- $|y_n - L| < \epsilon \forall n \geq N_2$

BS - Page 80 Exercise 7

Establish the convergence and find the limits of the following sequences

- $\left(1 + \frac{1}{n^2}\right)^{n^2}$
- $\left(1 + \frac{1}{2n}\right)^n$
- $\left(1 + \frac{1}{n^2}\right)^{2n^2}$
- $\left(1 + \frac{2}{n}\right)^n$

We know $\left(1 + \frac{1}{k}\right)^k$ converges to e

- We have $\{x_n\} = \left(1 + \frac{1}{n^2}\right)^{n^2}$

Let $\{x_k\} = \left(1 + \frac{1}{k}\right)^k$

Then $\{x_n\}$ is a subsequence of $\{x_k\}$, with the indices being the perfect squares,

Since we know $\{x_k\}$ converges to e , we also have that its subsequence $\{x_n\}$ converges to e

- We have $\{x_n\} = \left(1 + \frac{1}{2n}\right)^n = \left(1 + \frac{1}{2n}\right)^{2n}^{\frac{1}{2}}$

Let $\{x_m\} = \left(1 + \frac{1}{2n}\right)^{2n}$

Similarly, we have $\{x_m\}$ as a subsequence of $\{x_k\}$, with the indices being multiples of 2, thus $\{x_m\}$ converges to e

Since $\{x_m\}$ converges to e , we have that $\{x_n\} = (\{x_m\})^{\frac{1}{2}}$ converges to $\sqrt{e}$

- We have $\{x_n\} = \left(1 + \frac{1}{n^2}\right)^{2n^2} = \left(1 + \frac{1}{n^2}\right)^{n^2}^2$

Let $\{x_m\} = \left(1 + \frac{1}{n^2}\right)^{n^2}$

From earlier, we have $\{x_m\}$ converges to e

Since $\{x_m\}$ converges to e , we have that $\{x_n\} = (\{x_m\})^2$ converges to e^2

- We have $\{x_n\} = \left(1 + \frac{2}{n}\right)^n = \left(1 + \frac{1}{\frac{n}{2}}\right)^{\frac{n}{2} \cdot 2}$

Let $\{x_m\} = \left(1 + \frac{1}{\frac{n}{2}}\right)^{\frac{n}{2}}$

We have $\{x_m\}$ converges to e , as when $k \rightarrow \infty$ in $\{x_k\}$, $m \rightarrow \infty$ in $\{x_m\}$

Since $\{x_m\}$ converges to e , we have that $\{x_n\} = (\{x_m\})^2$ converges to e^2

BS - Page 80 Exercise 11

Suppose that $x_n \geq 0$ for all $n \in \mathbb{N}$ and that $\lim((-1)^n x_n)$ exists. Show that $\{x_n\}$ converges

Let $\lim((-1)^n x_n) = x$

$$\Rightarrow \forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that } |(-1)^n x_n - x| < \epsilon \forall n \geq N$$

Then for any even $k, k \geq N$

$$\Rightarrow |(-1)^k x_k - x| < \epsilon$$

$$\Rightarrow |x_k - x| < \epsilon$$

Also for any odd $p, p \geq N$

$$\Rightarrow |(-1)^p x_p - x| < \epsilon$$

$$\Rightarrow |x_p + x| < \epsilon$$

Since $x_n \geq 0 \forall n$ and $\epsilon > 0$

$$\Rightarrow x \geq 0, \quad x \leq 0$$

Because in the first equation if $x < 0$ it implies $x_k - x > 0$ meaning you could pick $\epsilon = \frac{-x}{2}$ to get a contradiction, and in the second if $x > 0$ implies $x_p + x > 0$ meaning you could pick $\epsilon = \frac{x}{2}$ to get a contradiction

Thus $x = 0$

Then

$$\Rightarrow \forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that } |(-1)^n x_n| < \epsilon \forall n \geq N$$

$$\Rightarrow x_n < \epsilon$$

Hence $\{x_n\}$ converges to 0